PERSONAL HOMELAB INFRASTRUCTURE

merox.dev Infrastructure Overview

Kubernetes · GitOps · Self-Hosted Cloud

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3	**30+**	**100%**	**~€0**	**<50'**
K8S NODES	SERVICES	GITOPS	CLOUD COST	FULL REBUILD

Table of Contents

1. Executive Summary 3

1. Executive Summary

This document covers the complete **merox.dev homelab infrastructure** — a production-grade self-hosted environment running on commodity hardware and free-tier cloud. Design philosophy: *declarative, reproducible, zero-cost cloud.*

Kubernetes Cluster

3-node Talos Linux HA cluster (all control-plane + worker) on Proxmox VMs. Managed 100% via FluxCD GitOps — a single `git push` deploys or updates any workload.

Oracle Cloud VPS

Always-free ARM VPS (4 vCPU / 24GB) hosting off-site services: reverse proxy, SSO, DNS, S3 backup storage, monitoring, and AI agent runtime.

Zero Open Ports

All external traffic via **Cloudflare Tunnel**. Management via **Tailscale** mesh VPN. No public IP, no open firewall ports required anywhere.

AI Automation

OpenClaw runs Claude-powered agents — manage infrastructure, publish blog posts, get daily briefings, all via Telegram.

2. High-Level Architecture & Network Topology

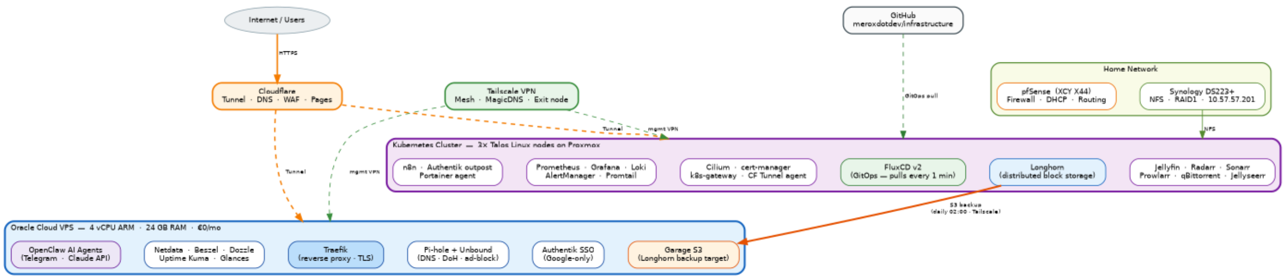

Figure 1 — Complete infrastructure topology: VPS ↔ K8s cluster ↔ home network, with Cloudflare Tunnel and Tailscale overlay

3. Hardware Inventory

Dell OptiPlex 3050 #1

K8s Node · Proxmox VM

CPU: Intel i5-6500T (4C/4T)
RAM: 16 GB DDR4
Storage: 128 GB NVMe
IP: `10.57.57.80`

Dell OptiPlex 3050 #2

K8s Node · Proxmox VM

CPU: Intel i5-6500T (4C/4T)
RAM: 16 GB DDR4
Storage: 128 GB NVMe
IP: `10.57.57.82`

Beelink GTi 13 Pro

K8s Node · Proxmox VM

CPU: Intel i9-13900H (14C/20T)
RAM: 64 GB DDR5
Storage: 2×2 TB NVMe
IP: `10.57.57.84` · iGPU i915

Dell PowerEdge R720

Proxmox Backup Server

CPU: 2× Xeon E5-2697v2 (24C/48T)
RAM: 192 GB DDR3
Role: PBS + DR VMs

Synology DS223+

NAS · NFS + Backup

Storage: 2×2 TB HDD RAID1
Protocol: NFS
IP: `10.57.57.201`
DB backups + media

XCY X44

Firewall / Router

CPU: Intel N100
RAM: 8 GB
OS: pfSense
Firewall · DHCP · Routing

Oracle Cloud ARM VPS

Off-site Cloud · Free tier

CPU: 4× vCPU ARM Ampere A1
RAM: 24 GB
Storage: 200 GB block
Cost: €0/mo

		Notes
		HA co
		Proxm
		Proxm
		Proxm
		NFS s
		Fixed
Portainer agent pod	10.57.57.103	Fixed IP via Cilium
	`100.72.22.38`	Management access only

4. Oracle Cloud VPS — Off-site Services

All services run as Docker Compose stacks under `/srv/docker/oracle-cloud/`. Traefik handles TLS. Authentik provides SSO.

Service	URL / Access	Purpose	Stack
Traefik	traefik.cloud.merox.dev	Reverse proxy · ACME TLS (Let's Encrypt) · routes all VPS traffic	traefik/
Authentik	sso.merox.dev	SSO identity provider — Google-only, no new signups	authentik/
Pi-hole + Unbound	pihole.cloud.merox.dev	DNS ad-blocking + recursive DoH resolver	compose.yml
Portainer EE	100.72.22.38:9000 (Tailscale)	Container management UI	compose.yml
Homepage	inside.merox.dev (Tailscale)	Internal dashboard — K8s + Proxmox + router	compose.yml
Joplin Server	joplin.cloud.merox.dev	Self-hosted notes sync (PostgreSQL backend)	compose.yml
Uptime Kuma	status.merox.dev	Service uptime monitoring + alerting	uptime-kuma/
Guacamole	rmt.merox.dev	Browser RDP/VNC/SSH gateway · Authentik SSO	guacamole/
Garage S3	garage.cloud.merox.dev	S3-compatible object store — Longhorn backup target	garage/
Netdata	netdata.cloud.merox.dev	Real-time metrics — parent + 3 child K8s nodes	compose.yml
Beszel	beszel.cloud.merox.dev	Lightweight host monitoring dashboard	beszel/
Dozzle	dozzle.cloud.merox.dev	Real-time Docker log aggregation UI	compose.yml
Glances	glances.cloud.merox.dev	System resource monitoring	compose.yml
OpenClaw Dashboard	agents.cloud.merox.dev	AI agent control panel (updated nightly)	agents-dashboard/
Code Server	code.cloud.merox.dev	Browser-based VS Code for remote editing	compose.yml

VPS Management Commands

```
cd /srv/kubernetes/infrastructure/vps

make health-check       # verify all services running
make setup              # full idempotent redeploy (Ansible)
make update             # OS package updates only
make check              # dry-run (--check --diff)
make restore            # interactive restore wizard (Joplin / Authentik)
make cleanup            # remove unused Docker images/volumes
make dr-full            # provision fallback VPS on Hetzner (~15 min)
```

5. Kubernetes Cluster — On-premise

3-node **Talos Linux** HA cluster — all nodes run control-plane + workloads (no dedicated workers). Managed via **FluxCD v2** GitOps from `meroxdotdev/infrastructure`.

Application Workloads

Infrastructure Services

Service	Namespace	Role
Cilium	kube-system	CNI · L2 LoadBalancer · Gateway API · NetworkPolicy
Longhorn	longhorn-system	Distributed block storage · S3 backup
cert-manager	cert-manager	Automated TLS (Let's Encrypt)
Cloudflare Tunnel	network	External access — zero open ports
k8s-gateway	network	Internal DNS for `*.merox.dev`
external-dns	network	Syncs K8s ingress → Cloudflare DNS
FluxCD v2	flux-system	GitOps controller
intel-device-plugin	kube-system	Exposes i915 iGPU for Jellyfin

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Observability Stack

Service	Namespace	Role
Prometheus	observability	Metrics + alerting rules
Grafana	observability	Dashboards
Loki	observability	Log aggregation
Promtail	observability	Log shipper (DaemonSet)
AlertManager	observability	Alerts + healthchecks.io
Portainer agent	default	Connects to Portainer EE (VPS)

```
roblem pods
lm releases + status
Flux sync

Talos config
e Talos version
e Kubernetes
```

6. Networking & Access Strategy

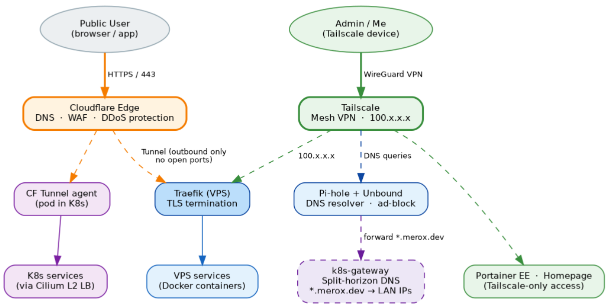

Figure 2 — Public vs management access paths

Cloudflare Tunnel (Public)

- Zero open firewall ports — anywhere
- All `*.merox.dev` public routes via tunnel
- TLS terminated at Cloudflare edge (free)
- DDoS protection + WAF included
- K8s: tunnel agent pod in `network` ns
- VPS: tunnel via Traefik container

Tailscale (Management)

- VPS + K8s nodes + Proxmox in same mesh
- MagicDNS for hostname resolution
- VPS acts as exit node
- Portainer EE & Homepage: Tailscale-only
- Longhorn → Garage S3 over Tailscale

Internal DNS (k8s-gateway)

Resolves `*.merox.dev` internally to Cilium L2 LB IPs — split-horizon DNS so internal clients use same URLs but traffic stays on LAN.

7. Storage & Backup Architecture

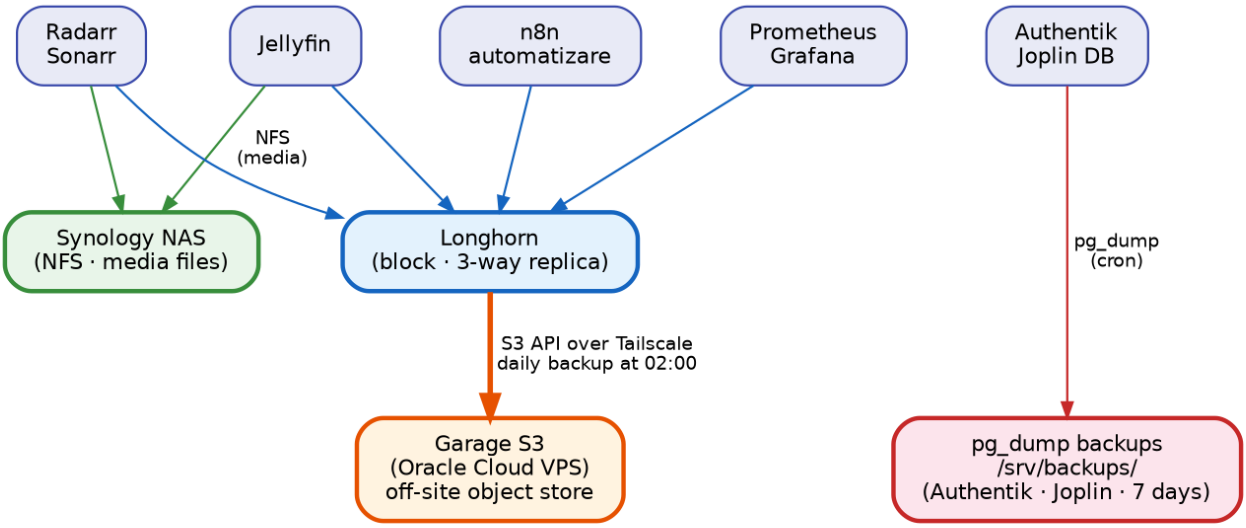

Figure 3 — Storage layers and backup data flows

CRITICAL — BACK UP AGE.KEY OFFLINE

`/srv/kubernetes/infrastructure/age.key` is the SOPS master key. Lost = all K8s secrets (Cloudflare token, encry

Storage Layers

Layer	Technology	Used For
Block (replicated)	Longhorn	App config, databases, state
File (NFS)	Synology DS223+	Media files, large datasets
Object (S3)	Garage (VPS)	Longhorn volume snapshots

Backup Schedule

What	When	Destination
Longhorn volumes	Daily 02:00	Garage S3 (VPS)
Authentik PostgreSQL	Manual pre-change	/srv/backups/authentik/
Joplin PostgreSQL	Automatic cron	/srv/backups/

8. GitOps & Automation Pipeline

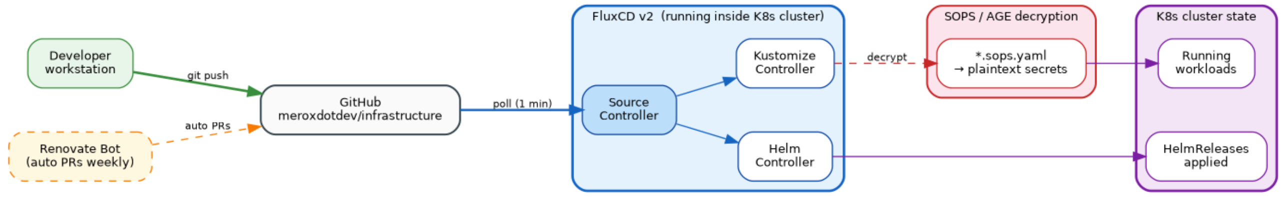

Figure 4 — FluxCD GitOps pipeline: git push → Flux reconciliation → K8s cluster state

SOPS Secrets

```
sops kubernetes/apps/<namespace>/<app>/app/secret.sops.yaml   # edit secret

# After rotating the AGE key
find . -name "*.sops.*" -exec sops updatekeys {} \;
```

FluxCD Pipeline

- Source Controller polls GitHub every 1 min
- Kustomize Controller applies `kubernetes/` manifests
- Helm Controller reconciles all HelmReleases
- SOPS/AGE decrypts secrets at apply time
- Drift detection — auto-reverts manual changes
- Force sync: `task reconcile`

Renovate Automation

- Runs every weekend via GitHub Actions
- Opens PRs for Helm chart + image tag updates
- Also tracks: Talos + Kubernetes versions
- Config: `.renovaterc.json5`
- Images tagged with `# renovate:` comments

Flux Troubleshooting

```
flux get sources git -A
flux get kustomizations -A
flux logs --level=error
flux reconcile kustomization cluster-apps --with-source

# HelmRelease stuck — suspend + resume
flux suspend helmrelease <name> -n <namespace>
flux resume  helmrelease <name> -n <namespace>
```

9. AI Agents — OpenClaw

OpenClaw runs Claude-powered agents on the VPS. Agents triggered via **Telegram bot** or cron. The `infra` agent can run `kubectl` and `docker` commands remotely in natural language.

Agent	Trigger	Purpose
news	Cron + Telegram	Daily briefing — HackerNews + RSS, summarised by Claude
blog	Telegram	Writes and publishes posts to merox.dev via GitHub Actions
infra	Telegram	Runs `kubectl` / `docker` — natural language infra management
costs	Telegram	Infrastructure cost tracking and reporting
design	Telegram	Visual content generation
orchestrator	Internal	Routes messages between agents + scheduled tasks
dashboard	Cron nightly	Updates agents.cloud.merox.dev
renovate	Internal	Git dependency sync + PR creation

10. Security Model

Secret Handling

Type	Method
K8s secrets	SOPS/AGE → `*.sops.yaml`
VPS Ansible secrets	Ansible Vault
Docker env vars	`.env` (gitignored)
Agent API keys	`.openclaw/.env` (gitignored)
Talos bootstrap	`talsecret.sops.yaml`

Access Control

Layer	Mechanism
External HTTP	Cloudflare Tunnel (no open ports)
Management	Tailscale mesh VPN
SSO	Authentik (Google-only)
K8s RBAC	Talos hardened defaults
Network policy	Cilium NetworkPolicy

11. Disaster Recovery

PREREQUISITES — HAVE READY BEFORE ANY DR SCENARIO

`age.key` (SOPS master) · Ansible Vault password · Tailscale auth key (check expiry!) · Cloudflare API token

DR Scenario Matrix

Scenario	Action	RTO
K8s cluster lost (nodes dead)	DR.md — provision DR VMs, bootstrap Talos, restore Longhorn from S3	~35 min
VPS lost (Oracle reclaims free tier)	`cd vps && make dr-full` then `make restore`	~15 min
Full rebuild from scratch	DEPLOY.md — Phase 1 (VPS) → Phase 2 (K8s) → Phase 3 (Agent)	~50 min
Single node lost	New Proxmox VM + `task talos:apply-node`	~10 min
New hardware (different IPs)	Edit `talconfig.yaml`, `cluster-vars.yaml`, `cilium/networks.yaml`	~20 min
HelmRelease stuck	`flux suspend/resume helmrelease`	<5 min

Full Rebuild — Phase 1 → 3

1. **Phase 1 — VPS (~15 min)**
   `cd vps && make dr-full` — provisions Hetzner VPS + deploys all Docker services via cloud-init. Then: `make restore` to restore Joplin + Authentik databases.
2. **Phase 2 — Kubernetes (~20 min)**
   Copy `age.key`. Edit `talos/talconfig.yaml` (node IPs, install disk) and `kubernetes/components/common/cluster-vars.yaml`.
   Run: `task bootstrap:talos` → `task bootstrap:apps` → `task restore:longhorn`
3. **Phase 3 — AI Agents (~15 min)**
   Create `openclaw` user. Run `claude login` (Anthropic OAuth). Run `openclaw onboard`. Fill `.openclaw/.env`. Enable systemd service.
4. **Validation**
   `kubectl get nodes` → all Ready · `docker ps | wc -l` → ~16 containers · Send Telegram message → bot replies.

```
# Useful during recovery
kubectl -n <ns> describe pod <pod>
kubectl -n <ns> logs <pod> --previous
talosctl -n <node-ip> health
talosctl -n <node-ip> dmesg
docker exec garage /garage status
```

12. External Dependencies & Costs

Service	Purpose	Cost	Tier
Cloudflare	DNS · CDN · Tunnel · WAF · DDoS · Pages (blog CI/CD)	€0/mo	Free
Tailscale	Management VPN mesh (all nodes + VPS)	€0/mo	Free
Oracle Cloud	Primary VPS — 4 vCPU ARM, 24GB RAM, 200GB	€0/mo	Always Free
Hetzner VPS	Fallback VPS — on-demand only if Oracle free tier lost	€5.39/mo	On-demand
Anthropic / Claude	Claude Pro — AI model powering OpenClaw agents	~$20/mo	Claude Pro
GitHub	Repos + GitHub Actions (blog CI, Renovate)	€0/mo	Free
Let's Encrypt	HTTPS certificates (auto-renew via cert-manager)	€0/mo	Free
Proxmox	Hypervisor for K8s VMs (own hardware)	€0/mo	Own HW
Synology DS223+	NAS — NFS + DB backups (own hardware)	€0/mo	Own HW

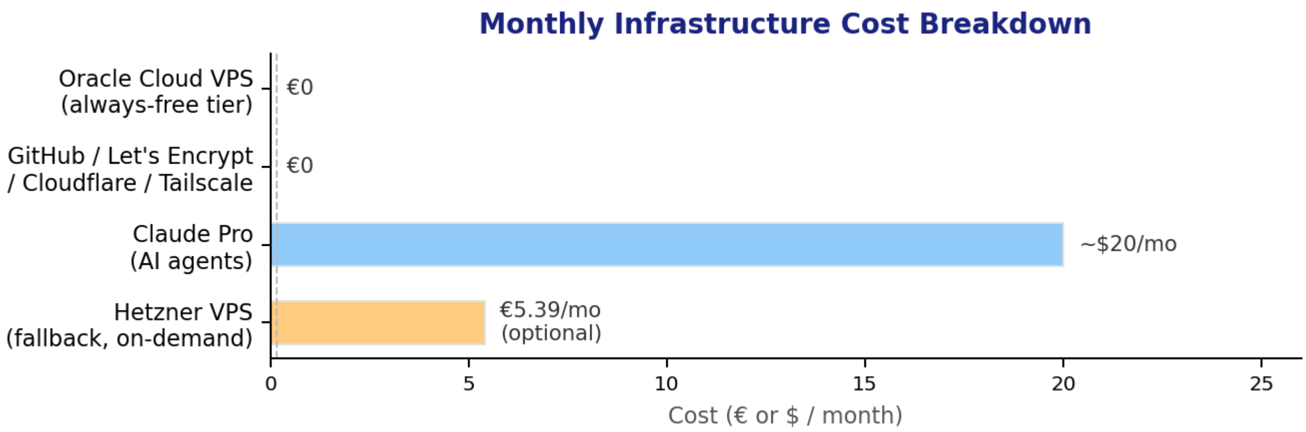

Figure 5 — Monthly cost breakdown (normal: ~$20 · with Hetzner fallback: ~€25)

Repository Map

What	GitHub Repo	Local Path
K8s Flux manifests + Talos config	meroxdotdev/infrastructure	/srv/kubernetes/infrastructure/
Ansible + Terraform VPS DR	meroxdotdev/infrastructure	/srv/kubernetes/infrastructure/vps/
Docker Compose VPS services	meroxdotdev/cloudlab-merox	/srv/docker/oracle-cloud/
OpenClaw agent config + skill	meroxdotdev/infrastructure	/srv/kubernetes/infrastructure/agent/
Blog (Astro)	meroxdotdev/merox (private)	/srv/merox/

13. Day-to-Day Operations Quick Reference

FULL DOCUMENTATION

Architecture index: `README.md` · Full rebuild: `DEPLOY.md` · K8s DR: `DR.md` · Jellyfin post-restore:

Kubernetes Cluster

```
kubectl get nodes
kubectl get pods -A | grep -v Running
```


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github.com/meroxdotdev


```

task talos:generate-config
task talos:apply-node   IP=10.57.57.80
task talos:upgrade-node IP=10.57.57.80
task talos:upgrade-k8s
```

VPS / Docker

```
cd /srv/docker/oracle-cloud
docker ps
```

```
make health-check
make setup
make update
make restore
make dr-full
```

Flux Troubleshooting

```
flux get sources git -A
flux logs --level=error
flux reconcile kustomization   cluster-apps --
with-source
flux suspend helmrelease <n> -n <ns>
flux resume  helmrelease <n> -n <ns>
```

Node Maintenance

```
kubectl drain <node>   --ignore-daemonsets   --
delete-emptydir-data
# do the work
kubectl uncordon <node>
# wait 1-2h between disk swaps
# for Longhorn replica rebuild
```

Pod Debugging

```
kubectl -n <ns> describe pod <pod>
kubectl -n <ns> logs <pod> -f
kubectl -n <ns> logs <pod> --previous
kubectl -n <ns> get events   --sort-
by='.metadata.creationTimestamp'
```

Longhorn + Garage S3

```
kubectl -n longhorn-system get volumes
kubectl -n longhorn-system get nodes.longhorn.io
# Remove orphaned replicas
kubectl get orphan   -n longhorn-system -o name
|   xargs kubectl delete   -n longhorn-system

docker exec garage /garage status
docker exec garage /garage bucket list
```

SOPS Secrets

```
sops kubernetes/apps/<ns>/<app>/  app/
secret.sops.yaml

# Rotate AGE key
find . -name "*.sops.*"   -exec sops updatekeys
{} \;
```